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DiFA: Inference-Time Forward-Process Alignment for Diffusion Models

Treating Each Denoising Step as a Kalman Observation Yields a Training-Free Temporal Consensus Wrapper with Significant FID Gains on CIFAR-10 and ImageNet

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Standard diffusion model inference treats each denoising step as an independent numerical integration, discarding the statistical correlation structure of predictions along the reverse trajectory. DiFA (**Di**ffusion **F**orward-process **A**lignment), accepted at ICML 2026, reframes inference as sequential state estimation inspired by Kalman filtering: each prediction is an observation of the true sample, fused with prior predictions according to noise-level compatibility to build a forward-aligned temporal consensus. The result is a training-free wrapper that wraps any existing sampler and yields consistent FID improvements.

AT A GLANCE

Problem: Inference treats denoising steps as independent estimators of x_0, ignoring forward-process statistical correlations across the trajectory.

Why it matters: Exploiting the statistical structure of the forward process during inference is free—it requires no training and can improve any existing sampler.

Method: Noise-level-weighted temporal consensus aggregates historical predictions; deviation guidance preserves fine-grained detail that averaging would smooth out.

Result: FID 4.67→3.21 on CIFAR-10 with DDIM (20 steps); FID 12.4→8.9 on ImageNet 256×256 with DDIM (50 steps).

Evidence: Ablations confirm both the consensus and deviation components are necessary; uniform averaging instead of noise-weighted degrades FID by 0.4–1.2.

The Big Picture

Diffusion models generate images by reversing a noise-injection process: starting from Gaussian noise x_T, a learned denoiser iteratively predicts the clean sample x_0 and steps toward it. The standard inference framework—DDIM, DPM-Solver, EDM—casts each step as deterministic numerical integration of the score function, treating the model's prediction $\hat{x}_0^{(t)}$ at each step as an independent, exact estimate.

DiFA identifies a missed opportunity: the sequence of predictions $\{\hat{x}_0^{(T)}, \hat{x}_0^{(T-1)}, \dots, \hat{x}_0^{(t)}\}$ is not independent—it represents a series of increasingly refined observations of the same unknown x_0, each corrupted by noise at level $\bar{\alpha}_t$. Kalman filtering is the optimal framework for fusing such a sequence of noisy observations. DiFA imports this framework into diffusion inference.

Why Existing Approaches Fall Short

The numerical integration perspective assumes each $\hat{x}_0^{(t)}$ is an unbiased, independent estimator of x_0. In practice:

Early predictions are structurally aligned but coarse. At high noise levels t, the model has little to work with—predictions are diffuse and uncertain. Yet they are statistically consistent with the forward process, making them useful priors for later steps.

Late predictions are fine but may drift. At low t, the model makes confident, detailed predictions—but these can deviate from the statistical manifold of the forward process, leading to artifacts or mode drops. Independent step treatment cannot detect or correct this drift.

No cross-step information reuse. Current samplers discard $\hat{x}_0^{(\tau)}$ for $\tau > t$ once step t is reached, wasting information that has already been computed.

Core Mechanism

DiFA introduces **Forward-Process Alignment (FPA)**: at step t, instead of using $\hat{x}_0^{(t)}$ directly, it computes a weighted consensus $\tilde{x}_0^{(t)}$ across all predictions to date:

$$\tilde{x}_0^{(t)} = \sum_{\tau \geq t} w(\tau, t)\, \hat{x}_0^{(\tau)}, \quad \sum_{\tau} w(\tau, t) = 1$$

Weights $w(\tau, t)$ are larger for predictions at noise levels close to t (structurally compatible) and smaller for predictions made at very different noise levels (less compatible):

$$w(\tau, t) \propto \exp\left(-\frac{\|\hat{x}_0^{(\tau)} - \hat{x}_0^{(t)}\|^2}{2\sigma^2}\right) \cdot \frac{1-\bar{\alpha}_\tau}{1-\bar{\alpha}_t}$$

To prevent over-smoothing, DiFA adds **Deviation Guidance**:

$$\hat{x}_0^{\text{DiFA}}(t) = \tilde{x}_0^{(t)} + \gamma\left(\hat{x}_0^{(t)} - \tilde{x}_0^{(t)}\right)$$

where $\gamma \in (0, 1)$ controls the trade-off between consensus smoothness and single-step detail.

Technical Formulation

The forward process is:

$$q(x_t \mid x_0) = \mathcal{N}\big(x_t;\, \sqrt{\bar{\alpha}_t}\, x_0,\; (1-\bar{\alpha}_t)I\big)$$

Treating each $\hat{x}_0^{(\tau)}$ as an observation of x_0 with additive Gaussian noise of variance proportional to $(1-\bar{\alpha}_\tau)$, the minimum-variance linear estimator is:

$$\tilde{x}_0^{(t)} = \left(\sum_{\tau \geq t} \frac{1}{1-\bar{\alpha}_\tau}\right)^{-1} \sum_{\tau \geq t} \frac{\hat{x}_0^{(\tau)}}{1-\bar{\alpha}_\tau}$$

DiFA approximates this by replacing the Gaussian noise assumption with the empirical prediction discrepancy, giving the noise-and-consistency weighted formula above. The two formulations agree when predictions are unbiased; DiFA's version is robust to the bias introduced by imperfect score estimation.

Learning or Inference Procedure

DiFA requires no additional training or fine-tuning. The implementation wraps the inner loop of any existing sampler:

1. Run the sampler's score evaluation to obtain $\hat{x}_0^{(t)}$.
2. Append $\hat{x}_0^{(t)}$ to the history buffer.
3. Compute $\tilde{x}_0^{(t)}$ via noise-level-weighted averaging over the buffer.
4. Apply deviation guidance to obtain $\hat{x}_0^{\mathrm{DiFA}}(t)$.
5. Pass $\hat{x}_0^{\mathrm{DiFA}}(t)$ to the sampler's step function in place of $\hat{x}_0^{(t)}$.

Memory overhead scales as $O(T \cdot H \cdot W \cdot C)$ for the history buffer; at 50 steps and 256×256 resolution, this is approximately 800 MB, which fits on a standard GPU.

What the Guarantee Says

Under Gaussian noise assumptions, DiFA's temporal consensus is the minimum-variance linear estimator of x_0 given all observations. This provides an optimality guarantee for the consensus itself.

For the full DiFA procedure (consensus + deviation guidance), the optimality guarantee applies in expectation to the forward-aligned component; the deviation term is a practical correction for the non-Gaussian case, empirically validated but without closed-form optimality.

Experimental Findings

CIFAR-10 unconditional (FID ↓):

Sampler	Steps	Baseline	+DiFA
DDIM	20	4.67	**3.21**
DPM-Solver++	20	3.44	**2.87**
EDM	35	2.04	**1.78**

ImageNet 256×256 class-conditional (FID ↓):

Sampler	Steps	Baseline	+DiFA
DDIM	50	12.4	**8.9**
DPM-Solver++	50	7.8	**5.6**

FD-DINOv2 (perceptual quality metric) and Inception Score improve consistently across all configurations. The largest absolute FID improvement occurs with DDIM, which makes independent step predictions with no built-in cross-step information reuse—confirming that DiFA fills a gap most pronounced in first-order samplers.

Ablations and Interpretation

Pure consensus ($\gamma = 0$): FID improves but visual sharpness drops; generated samples appear over-smoothed compared to reference images.

No consensus ($\gamma = 1$): Equivalent to baseline. The consensus is doing the work.

Full history vs. last 5 steps: Using the full prediction history improves FID by an additional 0.3–0.8 points, confirming that early-trajectory predictions carry useful structural information even when fused with weights that naturally down-weight them.

Uniform averaging vs. noise-level weighting: Replacing the noise-level compatibility term with uniform averaging degrades FID by 0.4–1.2 points. Noise-level weighting is critical: predictions made at very different noise levels have structurally incompatible biases that uniform averaging cannot account for.

Computational overhead: Adding DiFA to DDIM (20 steps) on a single A100 increases wall-clock inference time by 6.4%—the dominant cost is the buffer read and weighted average, not additional neural function evaluations.

Reference

Shigui Li, Delu Zeng. **DiFA: Inference-Time Forward-Process Alignment for Diffusion Models**. arXiv:2607.17972. ICML 2026. July 19, 2026. https://arxiv.org/abs/2607.17972